

ECON 100B: Macroeconomics

Class 6: The Solow Model 1 (Jones Chapter 5)

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Agenda

- Announcements
- Current events
- Finish up Chapter 4: A Model of Production
- Chapter 5: The Solow Growth Model

Announcements (same from Class 05)

- Plagiarism+ Quiz is due on bCourses this Friday 2/6
- Problem Set 1 is due on Gradescope on Friday 2/13
- Looking ahead:
 - Problem Set 2 will be due on Gradescope on Friday 2/27
 - In-class Midterm the next Thursday 3/5
 - Take-home Midterm Essay released 3/8, due 3/20
 - Spring Break 3/23-3/27

Read both of these newspapers

- Wall Street Journal
- New York Times
- For access to both, see [this Library guide](#)
- You might skip the op-ed pages unless you like being horrified or vindicated, depending on your point of view
- But sometimes there are very good op-ed pieces in both

Current events

- David Autor (MIT), Anton Korinek (U-VA), Natasha Sarin (Yale) (2026)
“[What if Labor Becomes Unnecessary?](#)” *New York Times* Round Table, Feb 4.
- Smart economists talking about AI, growth, and jobs
- A different take than Erik Brynjolfsson (Stanford) would probably have about the job market: Autor and Sarin remark that the timing of AI overlaps with macroeconomic slowdown, so job losses aren't obvious
- Korinek remarks that investment spending on AI is huge, and it also is directed toward firms that have very high market valuation per employee
- **Market valuation** is the value of the publicly traded capital stock. These measures are comparable to the capital labor ratio K/L like we saw in the Class 05 slide deck

First half of ECON 100B

- Chapter 2 shows how we measure income, $Y = C + I + G + NX$
- **Chapters 3–6 discuss long-run growth**
 - Chapter 3 introduced the growth rate rules

$$g[X \times Y] = g[X] + g[Y]$$

$$g[X/Y] = g[X] - g[Y]$$

$$g[X^a \times Y^b] = a \cdot g[X] + b \cdot g[Y]$$

- Chapter 4 introduces Cobb-Douglas production

$$Y = A K^{1/3} L^{2/3}$$

with constant returns to scale
and declining marginal productivity of K or L

- **Chapter 5 develops the Solow Model**
- Chapter 6 develops the Romer Model

Learning objectives in Chapter 4

- There are enormous differences in GDP per person, y , across countries
- The Cobb-Douglas production function $Y = A K^\alpha L^{1-\alpha}$ can help us model GDP. GDP per worker is $y = Y/L$
- This production function is realistic
 - Constant returns to scale:
If you duplicate a factory and its workers, you duplicate output
 - Diminishing returns to any one factor:
Increasing capital without increasing workers leaves idle capital
- What is technology, A ? If we assume it's the same across countries, we find a striking result: The model explains some of reality but far from all. Later, we will need to understand A

Part 2 of Chapter 4: Explaining GDP across countries

- How should we do this? It's challenging
- Suppose we measured Y , K , L , and inferred A for each country
... congratulations, you described reality. Now what?
- Well, maybe we could examine A across countries and see how crazy it looks. As a residual, it is true by definition. But is it crazy?
- What does “crazy” mean here? Something that is **orders of magnitude different** between otherwise similar countries on earth

Side note: Constant returns to scale provides us a nifty trick

- It turns out that $\frac{Y}{L} \equiv y = \frac{A K^{1/3} L^{2/3}}{L} = A k^{1/3}$ for $k \equiv \frac{K}{L}$

in words:

Income per worker y is a concave function of capital per worker k

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in words:

Income per worker y is a concave function of capital per worker k

- Now use your **growth rate rules**

$$g[y] = g[A] + \frac{1}{3}g[k]$$

- If we see a country with 6% less capital per worker, we would expect that its income per worker is 2% less ... plus any effect of total factor productivity (TFP)

Without TFP differences, the model doesn't do well

TABLE 4.3

The Model's Prediction for Per Capita GDP (U.S. = 1)

Country	Observed capital per person, $\bar{k}$	Predicted per capita GDP $y = \bar{k}^{1/3}$	Observed per capita GDP
United States			
Switzerland			
Japan			
Italy			
Spain			
U.K.			
Brazil			
China			
South Africa			
India			
Burundi			

Here, $\bar{k}$ for each country is divided by the U.S. level

So if the number you see is 0.914, that means capital per worker is 91.4% of the U.S. level

And a model with fixed $A = 1$ predicts $y = 0.914^{1/3} = 0.971$ or 97.1% of the U.S. level

Predicted per capita GDP is computed as $\bar{k}^{1/3}$, that is, assuming no differences in productivity across countries. Data correspond to the year 2019 and are divided by the values for the United States.

Source: Penn World Table, Version 10.0.

Without TFP differences, the model doesn't do well

TABLE 4.3

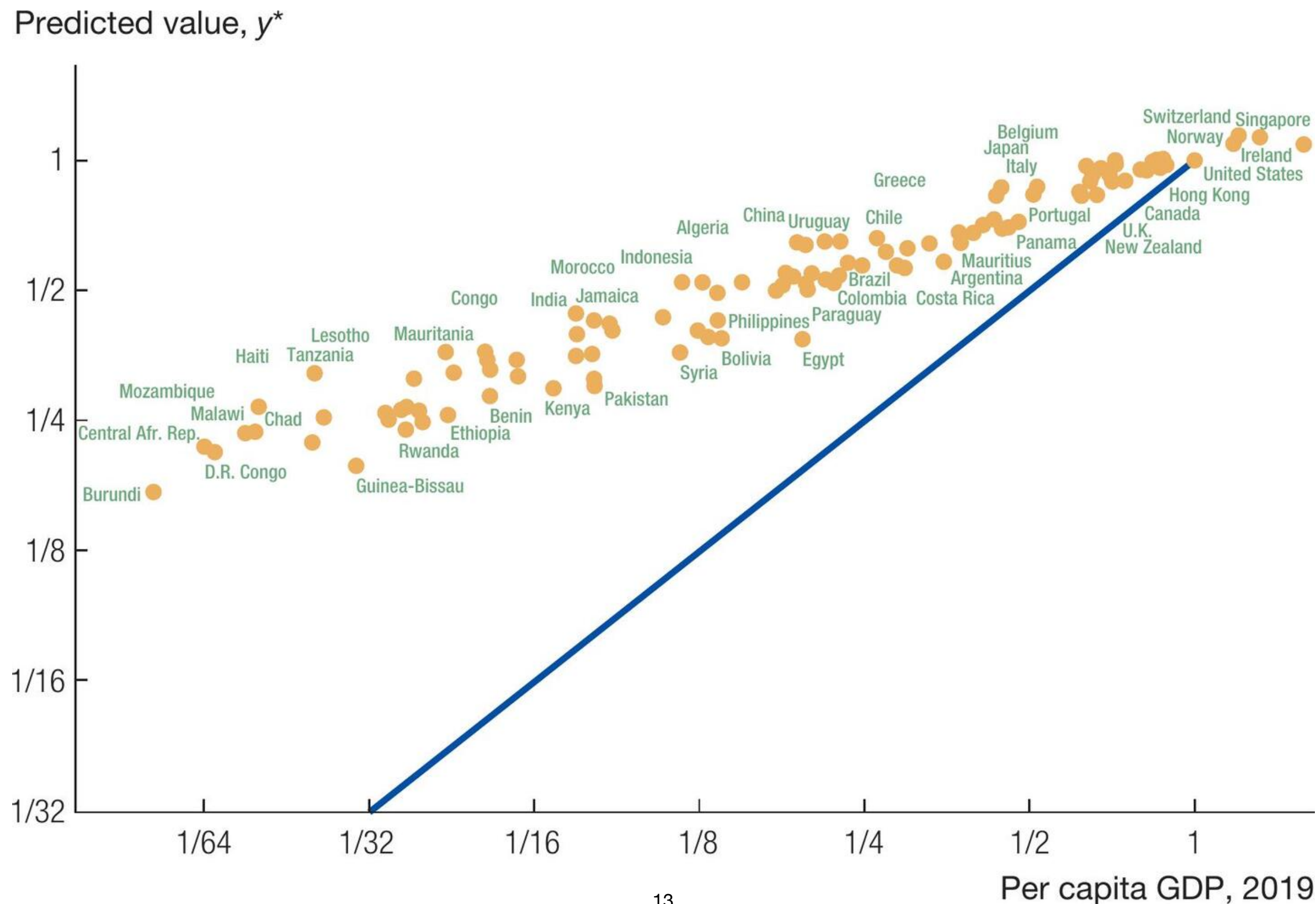
The Model's Prediction for Per Capita GDP (U.S. = 1)

Country	Observed capital per person, $\bar{k}$	Predicted per capita GDP $y = \bar{k}^{1/3}$	Observed per capita GDP
United States	1.000	1.000	1.000
Switzerland	1.488	1.142	1.203
Japan	0.914	0.971	0.634
Italy	0.792	0.925	0.651
Spain	0.716	0.895	0.645
U.K.	0.714	0.894	0.707
Brazil	0.195	0.580	0.233
China	0.275	0.650	0.226
South Africa	0.165	0.548	0.200
India	0.082	0.434	0.107
Burundi	0.005	0.172	0.013

Predicted per capita GDP is computed as $\bar{k}^{1/3}$, that is, assuming no differences in productivity across countries. Data correspond to the year 2019 and are divided by the values for the United States.

Source: Penn World Table, Version 10.0.

Without TFP differences, most predictions undershoot reality. But Switzerland, Ireland, Singapore are overperformers



When we compute TFP as the residual, we see big differences across countries

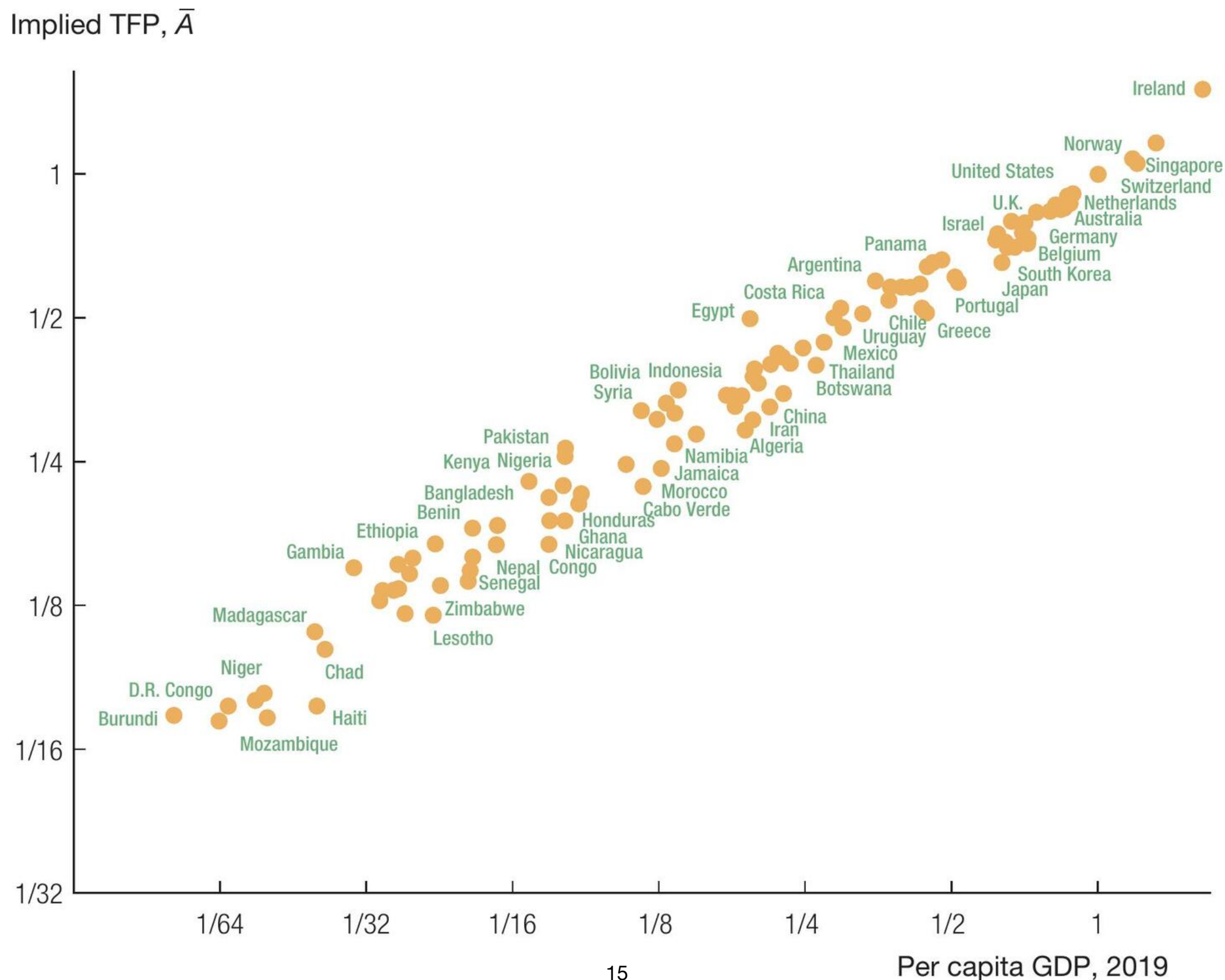
TABLE 4.4

Measuring TFP So the Model Fits Exactly

Country	Per capita GDP (y)	$\bar{k}^{1/3}$	Implied TFP ($\bar{A}$)
United States	1.000	1.000	1.000
Switzerland	1.203	1.142	1.054
U.K.	0.707	0.894	0.792
Japan	0.634	0.971	0.654
Italy	0.651	0.925	0.703
Spain	0.645	0.895	0.721
Brazil	0.233	0.580	0.402
South Africa	0.200	0.548	0.365
China	0.226	0.650	0.347
India	0.107	0.434	0.247
Burundi	0.013	0.172	0.074

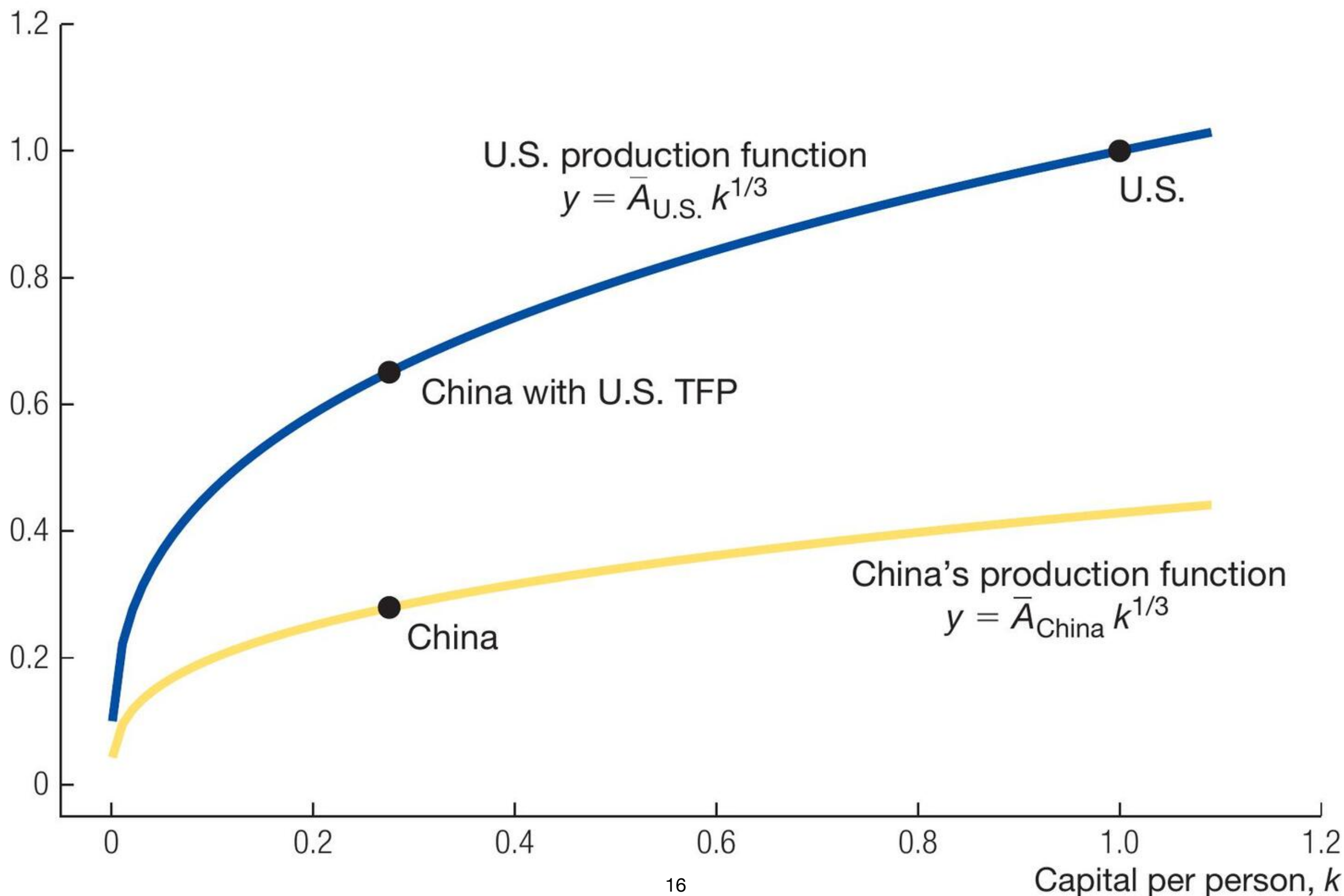
Calculations are based on the equation $y = \bar{A}\bar{k}^{1/3}$. Implied productivity $\bar{A}$ is calculated from data on y and $\bar{k}$ for the year 2019, so that this equation holds exactly as $\bar{A} = y/\bar{k}^{1/3}$.

TFP across countries is a “luxury good,” rising strongly with income per person



TFP matters a lot. It shifts the production function upward, raising output per person as a function of capital per person

Output per person, y



Teaser for Chapter 6 and Nobel Prize

Good income measures for North Korea are not readily available. However, this image of nighttime light output vividly conveys the relative income differences between North and South Korea. The obvious explanation is institutional differences. But which institutions, exactly?



Like a Barnaby?

Best [mondegreen](#) ever?



Learning objectives in Chapter 5

- Investment in new capital comes from **saving**, which usually comes from domestic income not consumed: $S = Y - C = I$
- If people work and save at rate s out of income, then $S = sY$
- But capital also breaks at rate d
- The Solow Model includes 2 curves: **saving** sY , and **depreciation** dK ,
and where they cross, $sY = dK$, capital is unchanging at K^*
in what we call a **steady state**
- When countries are far below their steady states, they **grow fast**
- A simple extension in §5.10 is a steady state with $k^* = \frac{K}{L}$ with
population growth at rate n

Buckle up, we're in for some chop

- Population usually grows. A Solow model with population growth at rate $g[L] = n$ is best
- But it's harder to understand
- A little calculus *and our growth rate rules* reveal the story in §5.10
- On exams, I expect you to be able to use the model, not derive it
- My slides proceed a little differently than the 6e textbook, more like the 1e. But underneath it all are the same pieces

Primitive farming as a simple example

- A family somehow farms corn with their hands. **People = labor**
- Each harvest, the family reaps their corn by hand. **Harvest = income**
- They're hungry, and they need to eat some corn
- But if they consumed all their corn, there wouldn't be any **seed corn** to plant and produce new corn at the next harvest
- So the family eats $c = 0.75$ of their corn and saves $s = 0.25$ of their corn, placing the saved corn in a silo for planting later
- All corn harvested is output = income, Y
- The eaten corn is consumption, C
- The saved corn is saving S that becomes investment I and new capital ΔK . Old capital is fully used up in production; depreciation = 100%

Translating this into our model

- The primitive farm economy produces output Y_t in year t using K_t capital in year t and L_t labor in year t via our production function:

$$Y_t = F(K_t, L_t) = A K_t^{1/3} L_t^{2/3}$$

- The output Y_t is corn and it is either consumed or invested:

$$Y_t = C_t + I_t$$

- Investment produces new capital. Capital in year $t + 1$ equals capital left over after depreciation in year t plus new investment

$$K_{t+1} = (1 - d)K_t + I_t$$

Interpretation and a little algebra

$$K_{t+1} = (1 - d)K_t + I_t$$

- For example, when $d = 0.10$, then $1 - d = 0.90$ and 90% of the capital stock persists into the next period, plus any new investment
- Distribute and rearrange

$$K_{t+1} = K_t - d K_t + I_t$$

$$K_{t+1} - K_t = -d K_t + I_t$$

$$\Delta K_{t+1} = I_t - d K_t$$

- Change in the capital stock is new investment minus depreciation

$$\Delta K_{t+1} = I_t - d K_t$$

- Suppose we start with $K_0 = 1000$, and each period, investment $I_t = 200$ and depreciation $d = 0.10 = 10\%$

time	K_t	I_t	$d K_t$	ΔK_t	K_{t+1}
0	1000	200	100	100	1100
1	1100	200	110	90	1190
2	1190	200	119	81	1271

Other elements of the model

- We'll assume that labor $L_t = \bar{L}$ is exogenously fixed
- What about the level of investment, I_t
- Like the farmers with the corn, we'll assume that agents save a fixed fraction s of income, which becomes investment

$$I_t = s Y_t$$

Equations in the model

$$Y_t = A K_t^{1/3} \bar{L}^{2/3}$$

$$L_t = \bar{L}$$

$$\Delta K_t = I_t - d K_t$$

Equations in the model

$$Y_t = A K_t^{1/3} \bar{L}^{2/3}$$

$$L_t = \bar{L}$$

$$\Delta K_t = I_t - d K_t$$

$$I_t = s Y_t$$

$$\Delta K_t = s Y_t - d K_t$$

Equations in the model

This is a concave
function of K

$s Y_t$

This is a linear
function of K

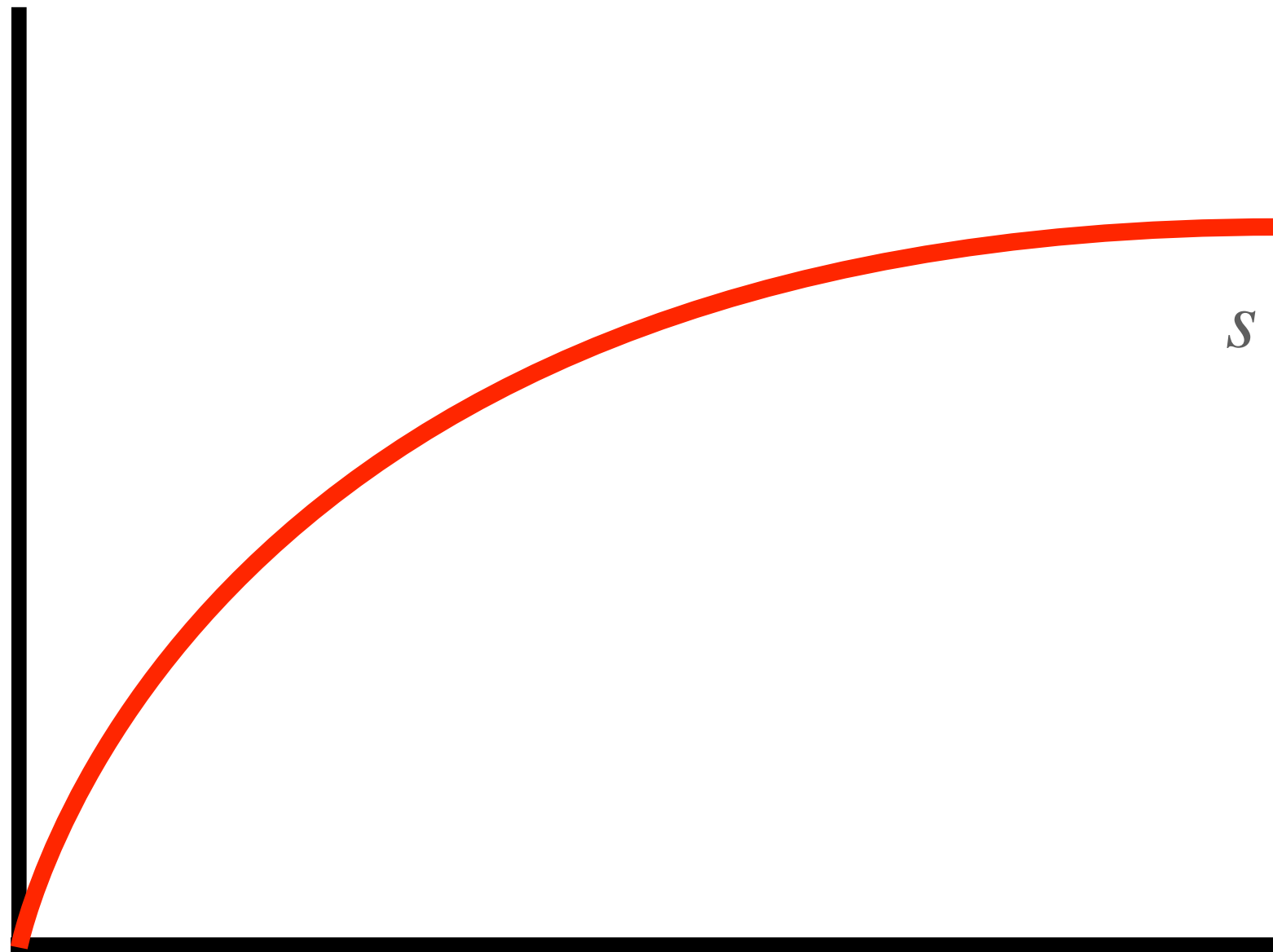
$d K_t$

$$\Delta K_t = s Y_t - d K_t$$

Let's look for
a crossing
where they
are equal

The saving curve $s Y_t$ is a concave function of capital K_t

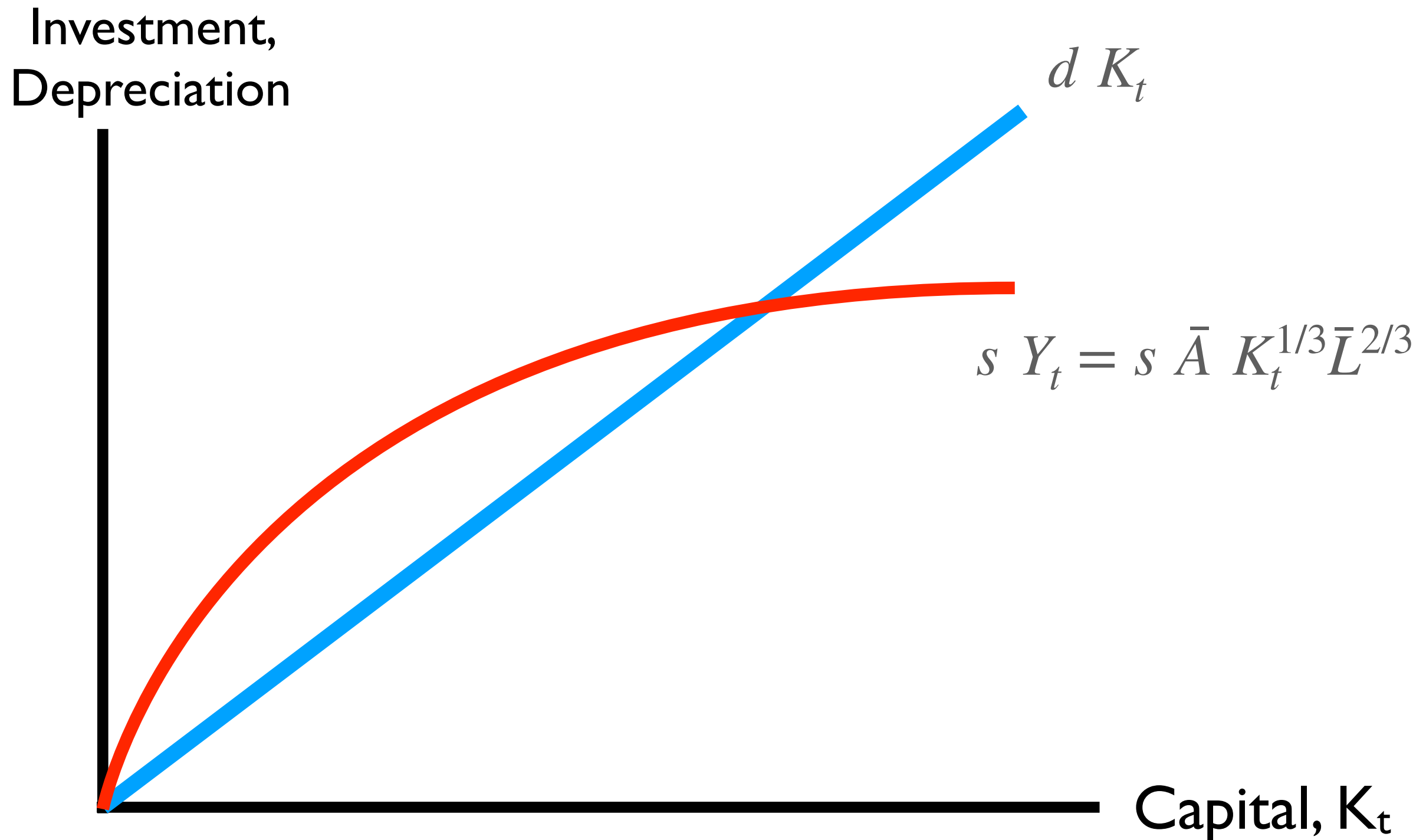
Investment,
Depreciation



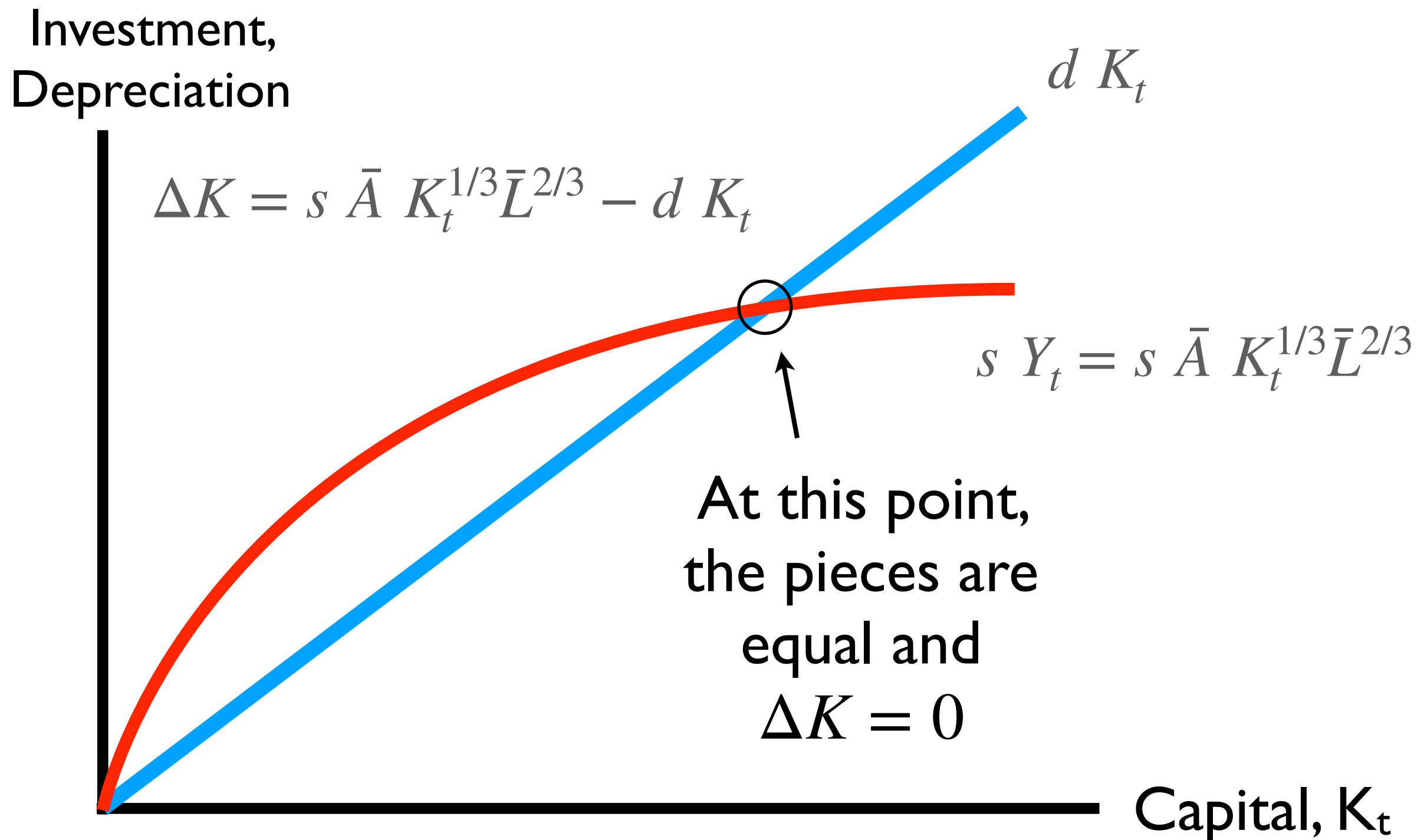
$$s Y_t = s \bar{A} K_t^{1/3} \bar{L}^{2/3}$$

Capital, K_t

The depreciation line is a straight line
function of capital K_t



The Solow diagram graphs these two pieces together, with K_t on the x-axis



The curves cross at the **STEADY STATE**

